

Unified Tool Surface and Extended Bag Relational Algebra

One immutable artifact system across Direct, Atomic, and Hybrid modes

DIAGNOSTIC ONLY

DIRECT = P + SQL + C

ATOMIC = P + R + C

HYBRID = DIRECT $\cup$ ATOMIC

Current experiment mode · SQL and atomic artifacts interoperate

SHARED PERCEPTION · P

 **describe_table**

Discover source schema

 **inspect_column**

Bounded value profile

 **read_rows**

Observation only; no artifact

Outputs: schemas and bounded observations

DIRECT SQL

 **execute_sql**

One read-only SELECT / WITH
Registered relation namespace
Top-level ORDER BY is explicit

 **SQL safety envelope**

Authorizer · timeout
canonical dynamic types
row / byte / cell limits

Produces relation artifact

ATOMIC OPERATORS · R

 **filter_rows**

rows only · preserve order

 **project**

columns / expressions

 **join**

binary · bag multiplicity

 **aggregate**

group and metrics

 **distinct**

all visible columns

 **set_operation**

aligned schemas

 **sort**

establish order

 **limit**

ordered input required

 **add_rank**

add ranking column

SHARED CONTROL · C

 **commit_checkpoint**

Meaningful milestone
summary · uncertainty
1–3 next targets

 **restore_checkpoint**

Explicit contradiction
restore exact snapshot
create recovery child

 **answer**

Consume exactly one active relation artifact
The exact full relation is the answer evidence

Zero checkpoints are legal for simple tasks

IMMUTABLE RELATION ARTIFACT

handle · kind · ordered columns and canonical types · row_count · ordered_by · Harness derivation
Every producer publishes atomically. Direct and atomic tools can consume the same active artifact.
Hidden ordering metadata never enters visible columns, Direct SQL, context, or answer.

F04 · Mode surfaces

$$\mathcal{I}_D = \mathcal{P} \cup \{\text{execute_sql}\} \cup \mathcal{C}, \quad \mathcal{I}_A = \mathcal{P} \cup \mathcal{R} \cup \mathcal{C}, \quad \mathcal{I}_H = \mathcal{I}_D \cup \mathcal{I}_A$$

F05 · Artifact contract

$$A = \langle h, k, \langle c_i : \tau_i \rangle_{i=1}^m, n, \omega, d \rangle$$

F06 · Extended relational algebra

$$\sigma_p(R), \quad \pi_{\langle e_i \rightarrow a_i \rangle_{i=1}^m}(R), \quad R \bowtie_{\theta} S, \quad \gamma_{G; \langle f_i(c_i) \rightarrow a_i \rangle}(R), \quad \tau_K(R)$$

F07 · Bag join multiplicity

$$\mu_{R \bowtie_{\theta} S}(r, s) = \mu_R(r) \mu_S(s) \mathbf{1}[\theta(r, s)]$$